

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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Abstract

How could activity distributed across cortex support one experienced situation? We separate three requirements: *coherence* (collective phase order), *compatibility* (local descriptions agree where their domains overlap) and *self-representation* (the state carries an accurate reconstruction of itself). Coherence supplies neither of the others, and five results establish that with no cortical hypothesis: synchrony constrains content only through encoders shared on overlaps, and uniformly over pairs only inside a patch too small to span the territories at issue; the resultant does not determine the state; a unique fixed point is not a self-model; where the encoding is a region's restriction, the region must resolve the relevant family before any readout is chosen; and agreement about a change takes communication. That grain is the uniform bound's, not phase order's: three results replace it by a count of the pairs that can disagree, a chain across a cover's shared sites, or a spectral gap at a frequency-locked configuration, each against a declared input coherence does not supply. Four are checkable for a machine candidate, fixing what a claim of machine unity commits its maker to measuring. Under stated hypotheses, compatible local states

glue to a unique global state that an accurate contracting reconstruction fixes uniquely; installed energy bounds a declared actuator’s mean coupling, so a coherent branch requires it to exceed $2D/\kappa$, for a κ derived here for neither cortex nor a processor. A Lean 4 development carries these results and their composition, with the eight connecting hypotheses as explicit arguments, no incomplete proofs and an audited axiom footprint. One prediction needs no absolute calibration: onset delay scales as the square root of the rate at which coupling crosses threshold, which a within-subject emergence-from-anaesthesia design can test. We propose the glued state and its internal reconstruction as correlates of unity and a minimal self, and an extracellular-field hypothesis linking awakening to rising coupling; testing it requires calibrated geometry, resolved phase dynamics and validated decoders.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Successful coordination would still leave a representational question: what makes the coordinated activity a description of one situation, with a perspective on that situation?

We distinguish three requirements. *Coherence* concerns collective phase order. *Compatibility* concerns whether local descriptions agree where their domains overlap. *Self-representation* concerns whether the resulting state carries an accurate internal reconstruction of itself. We propose the latter

14 two relations as formal correlates of conscious unity and a minimal self; their
15 interpretation requires evidence beyond the phase model.

16 Unity has been approached differently elsewhere: integrated information
17 theory treats integration and exclusion as constitutive (Tononi et al., 2016),
18 global-workspace accounts tie availability to broadcast (Baars, 1997; Dehaene
19 et al., 2014), and predictive-processing accounts derive both from a hierar-
20 chical generative model (Friston, 2010). Here the three requirements are
21 kept apart rather than derived from one another, so that each carries its own
22 assumptions and admits its own measurement; Section 10.3 returns to the
23 comparison.

24 Consider reaching for a cup. A visual description of its location and a
25 description of the hand’s relation to it can agree about where the cup is while
26 contributing different contents. Synchronizing the two processes would leave
27 disagreement about that shared quantity unresolved. An internal representa-
28 tion must additionally locate the perceiver within the represented situation.
29 This example motivates the requirements; the mathematical constructions
30 use spatial measures and fields in real normed vector spaces, whose identifi-
31 cation with perceptual content is a further modelling task.

32 The mathematical contribution is to specify connections between these
33 requirements and the assumptions each connection needs. A declared ac-
34 tuator’s stored energy bounds its mean coupling. Phase encoders shared on
35 overlaps convert phase order into a bound on content disagreement. Compat-
36 ible spatial measures have a unique global extension; approximate agreement

controls a specified global selection. A contracting reconstruction makes that extension its unique fixed point only if it accurately reconstructs the extension. Accuracy across a family of states imposes a further condition on the information retained by the encoding.

The empirical contribution is to identify what observations can decide. Stationary phase distributions can be shared by stronger coupling, weaker diffusion and a matched common drive. Finite ramps change the observable onset. A winding field is stationary at the same coupling as a uniform one and has the opposite resultant, so that observable does not determine the state. Coherent adaptation can lose alignment with a designated environmental template, and a compatibility estimator needs diagnostics that distinguish content from decoder artifacts. These findings determine the measurements needed to test the proposed connections.

Three of the ingredients are standard: the synchronization threshold $K_c = 2D$ for the noisy mean-field model (Sakaguchi, 1988), the unique gluing of compatible local sections, and the Banach fixed-point theorem (Banach, 1922). Four things here are not. The installed-energy condition makes coupling a purchased quantity: a declared actuator’s stored energy bounds the mean coupling it supplies, so coherence on the stationary branch carries a resource requirement in units a measurement can reach, and for cortex that requirement reduces to a figure the mode prices must deliver. The coherence-to-content bound converts phase order into a statement about descriptions rather than about timing, and its patch-local form is what keeps it from

60 being vacuous in a large population. A stability criterion on the reweighted
61 coupling kernel, together with an existence condition on an amplitude layer,
62 settles when a low resultant reports weak coupling and when it reports a
63 winding field. And the composition is machine-checked together with a wit-
64 ness discharging all eight hypotheses at once, which is what makes their
65 conjunction non-empty rather than a list no object satisfies.

66 Granting those premises, collected in Table 1, the account runs as follows.
67 Extracellular contraction on awakening raises endogenous field amplitude
68 and with it the effective coupling between cortical oscillators. Once that
69 coupling passes $2D$, a coherent phase branch exists, and from an initial spread
70 within a quarter turn the population settles onto it; other initial data leave
71 windings stationary at the same coupling. Encoders shared where cortical
72 territories overlap turn that phase order into agreement about the quantities
73 the territories describe in common, so the local descriptions glue: there is
74 one state of which each is a restriction. Part of that state encodes the state
75 itself, and a readout of that part reconstructs it accurately enough, and
76 sensitively enough across the states that matter, to make it the unique state
77 the system models itself as being. On this account the glued state is what
78 makes distributed activity one situation, and its accurate self-reconstruction
79 is what gives that situation a perspective. Every step of it is conditional.

80 Section 2 collects the results that hold whatever physical field is proposed
81 and whatever substrate carries it; a reader who rejects the cortical hypothesis
82 of Section 9 still has to accommodate them. Sections 3–5 then develop the

83 three requirements together (Fig. 1), and Section 6 reads the unconditional
84 results against a machine candidate. Section 7 composes them, stating the
85 thermodynamic constraints and all eight connecting hypotheses per branch;
86 its supporting models of feedback, learning and resource supply have distinct
87 state spaces and assumptions. Section 8 then examines what the observables
88 can establish, before Section 9 develops the cortical hypothesis and an awak-
89 ening experiment. The hypothesis is that extracellular contraction increases
90 field-mediated coupling; its direction, rate and relation to recovery require
91 independent calibration. Supplemental Material organizes the constructions
92 and controls behind the claims they support. Table 2, at the end, lists the
93 symbols and the section that defines each.

94 **2 What holds without the cortical hypothesis**

95 Most of what follows is conditional. Table 1 names eight hypotheses that
96 the composition consumes rather than derives, and the cortical identification
97 of Section 9 is a further commitment on top of them. Five results are not
98 conditional in that way. They constrain any account that reads unity off
99 phase coherence, whatever physical field is proposed to carry the coupling
100 and whatever substrate carries the field. Each is stated below where it is
101 used; collected here, they are the article's fixed points.

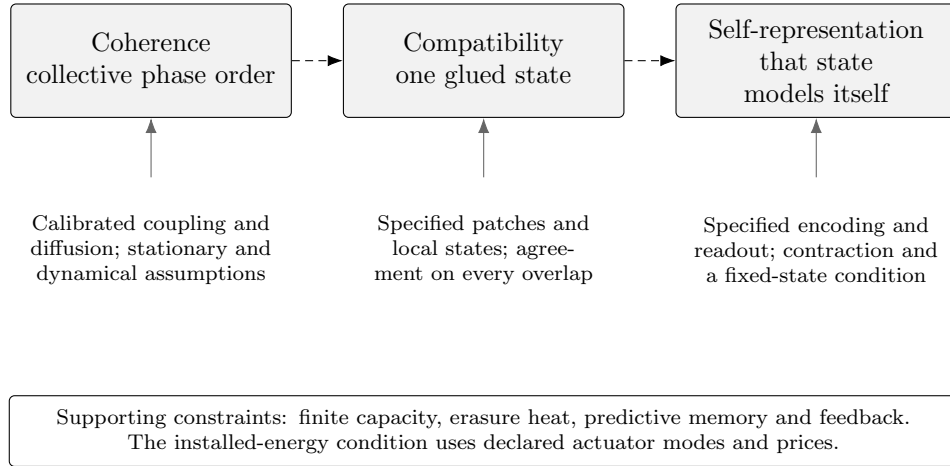


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

102 **Phase order constrains content only inside a patch too small to**
 103 **carry unity.** Assign every patch the same phase and let two local mass
 104 profiles disagree at a shared site. The profiles are chosen independently of
 105 phase, so synchrony places no constraint on their disagreement, and they
 106 have no common extension: perfect locking is consistent with local descrip-
 107 tions that do not glue (Section 4). What the positive result of Section 4.2
 108 costs is exactly this gap. A bound on content disagreement needs encoders
 109 already related on overlaps, and that relation is a supplied fact about the de-
 110 scriptions rather than a consequence of the dynamics. Supplying it relocates
 111 the limit rather than lifting it. Eq. (6) bounds every pair at once and a chord

112 never exceeds 2, so at the patch-local order of the sheet in Section 8.3 that
 113 uniform reading carries information only at 3 sites or fewer. The grain is the
 114 uniform reading's and not phase order's, and Section 4.2 states three results
 115 without it: a scale-free count of the pairs that can disagree, which names
 116 none of them; a bound chained through the sites a cover's patches share,
 117 which travels a distance linear in the hops taken; and, at a frequency-locked
 118 configuration, a bound read off the coupling graph's algebraic connectivity,
 119 which carries no population count at all. What none of the three changes is
 120 the constraint stated here: a count locates no site, and the other two con-
 121 sume a declared cover or declared hardware data, so phase order by itself
 122 still says nothing about territories too far apart to share a patch, which is
 123 where the compatibility claim is made.

124 **The order parameter does not determine the state.** A field wind-
 125 ing once across a periodic sheet has resultant $r = 0$ — the order parameter
 126 of Eq. (1) below — while every patch of it stays locked, and this is not a
 127 contrived initial condition. On a periodic array of any dimension, a winding
 128 is stationary for *any* isotropic symmetric kernel at identical frequencies, be-
 129 cause the coupling term is odd in the separation against a kernel even in it
 130 (`char_is_kuramoto_trajectory`). Section 8.3 exhibits a sheet on which uni-
 131 form and winding initial conditions reach 0.9657 and 0.0083 at one coupling,
 132 diffusion and kernel. A low resultant therefore reports either weak coupling
 133 or a winding field, and no measurement in r alone separates them. The state

134 also carries an integer the resultant does not: read around a ring, the same
 135 winding has winding number q , and a nonzero q obstructs every global real-
 136 valued phase lift (`winding_degree_obstructs`). The resultant is an average
 137 over sites and moves when they are reweighted or the patches redrawn; the
 138 degree telescopes and does not.

139 **A unique fixed point is not a self-model.** A constant readout con-
 140 tracts, fixes whatever state it returns, and reconstructs nothing. Uniqueness
 141 is consequently the wrong criterion, and Section 5.2 gives the right shape
 142 of one: reconstructing M states pairwise more than 2ε apart within error ε
 143 requires M distinguishable codes, so an alphabet of 2^b codes caps the family
 144 at $M \leq 2^b$. This is a counting argument and holds for any encoder and
 145 readout whatever.

146 **Where the encoding is a restriction, the region must resolve the**
 147 **family.** Read the code as the state's own restriction to a region and
 148 the criterion moves off the readout entirely. A boundary whose region
 149 reads two relevant states identically cannot reconstruct them more than 2ε
 150 apart (`not_isRestrictionResonance_of_reconstructs`), and a resonant
 151 boundary that does reconstruct a separated family restricts injectively on
 152 it (`restrictToAvatar_injOn_of_isRestrictionResonance`). Where the
 153 region sits is fixed before an encoding is chosen, so no readout supplies
 154 resolution the region lacks.

155 **Agreement after a change takes communication.** In a synchronous
156 network whose updates read only the site’s own state and its declared neigh-
157 bours’ messages, two configurations agreeing on a site’s causal past through
158 T rounds leave that site identical at round T . Two interventions outside
159 that past that require reports more than 2ε apart therefore cannot both be
160 reported from it within ε by the deadline (Section 5). Snapshot compatibil-
161 ity and timely compatibility are therefore different requirements, and cutting
162 the path removes the second while leaving the first intact.

163 None of these depends on the coupling being electromagnetic, on the sub-
164 strate being cortex, or on the eight hypotheses of Table 1 holding. Section 6
165 applies four of them — all but the one about the resultant, which is a state-
166 ment about a spatial observable — to a candidate that is neither biological
167 nor field-coupled.

168 **3 Coherence: physical coupling and its re-** 169 **source condition**

170 **3.1 Dynamics and the stationary threshold**

171 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

172 Here θ_i is the phase of oscillator i , ω_i its natural frequency, $D > 0$ is phase dif-
 173 fusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units,
 174 with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; fre-
 175 quency heterogeneity and diffusion are distinct parameters. The stationary
 176 calculation below uses identical frequencies and this scalar coupling. Ap-
 177 plying it to a spatial kernel requires an aggregation rule and a check of the
 178 resulting approximation.

179 For identical frequencies in their rotating frame, the positive normalized
 180 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

181 where I_n are modified Bessel functions and ψ is the collective phase. The for-
 182 mal derivation uses the absence of stationary current for a periodic gradient
 183 drift to obtain this density, then verifies normalization and self-consistency.
 184 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988). Above
 185 threshold, exactly one positive solution coexists with zero and increases
 186 strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

187 Figure 2 shows this stationary branch.

188 Linearization about the uniform density gives rate $K/2 - D$ for the first
 189 harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists

190 exactly above threshold. About the coherent density, the quadratic form
 191 is strictly dissipative transverse to collective rotation, whose rate is zero.
 192 These linear estimates concern the limiting equation. Relating them to finite
 193 stochastic trajectories requires a dynamical mean-field limit and branch selec-
 194 tion; exchange symmetry alone does not establish the needed independence
 195 (Sznitman, 1991).

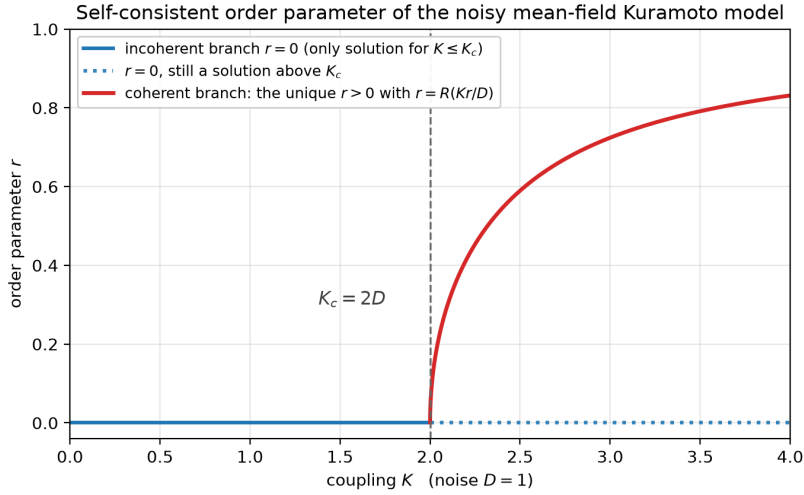


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

196 3.2 Installed energy bounds mean coupling

197 A spatial model replaces the finite sum by integration against a kernel. One
 198 declared implementation carries response modes with continuous profiles ϕ_i ,

199 installation prices u_i and nonnegative occupancies $c_i(\Xi)$ in configuration Ξ :

$$K_{\Xi}(x, y) = \sum_i c_i(\Xi) \phi_i(x) \phi_i(y), \quad U(\Xi) = \sum_i u_i c_i(\Xi). \quad (4)$$

200 The same configuration sets the reciprocal kernel and stored energy. An
 201 executed transition changes the kernel through its changed occupancies and
 202 requires work $U(\Xi') - U(\Xi) + Q$, with Q positive into the reservoir. The
 203 installed kernel is averaged over the resulting configuration law $p(\Xi)$. Cell-
 204 pair reconstruction on refining spatial partitions converges to its product-
 205 measure integral under the specified compactness and sampling assumptions;
 206 the refinement index is independent of the number of control updates.

207 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 208 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's
 209 integral gives

$$\begin{aligned} U_{\text{inst}} &= \sum_{\Xi} p(\Xi) U(\Xi), \\ K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_{\Xi} p(\Xi) \sum_i c_i(\Xi) m_i^2 \leq \kappa U_{\text{inst}}. \end{aligned} \quad (5)$$

210 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 211 coherent stationary branch and any growing Fourier mode about the uniform
 212 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 213 is necessary: energy above it need not produce sufficient coupling because
 214 the bound can be loose.

215 The quantity constrained is installed energy rather than dissipated heat,
 216 and the condition binds through κ rather than through the energy, since
 217 any macroscopic substrate carries installed energy far above the thermal
 218 floor. Read forwards it therefore excludes no macroscopic candidate. Read
 219 backwards it is a number. Take the region the field arithmetic takes, a
 220 sphere of one LFP decay length holding 1676 neurons, at the metabolic rates
 221 Supplemental Material collects and over one phase-diffusion time $1/D = 2$ s
 222 at $D = 0.5$ rad²/s: the installed energy is 1.0×10^{-6} J, and the condition is
 223 met at every κ above 9.6×10^5 J⁻¹s⁻¹. That figure is what an identification
 224 of cortical mode prices has to deliver, and the first identification delivering
 225 less refutes the coupling-gated account for cortex at this region and interval.
 226 Another interval rescales the figure by its ratio to this one.

227 What the condition discriminates, it discriminates through that factor.
 228 The required κ scales as D^2 and inversely with the region's cell count, so at
 229 fixed mode prices the condition ranks candidate substrates by their installed
 230 energy. In a declared implementation it decides outright, the learning witness
 231 of Section 7 clearing the line after one update while its matched control stays
 232 on the exclusion side at the same conversion factor and diffusion. What is
 233 missing for cortex is a price list rather than a mechanism. A mode is a priced
 234 rank-one contribution to the kernel, cortical field generation has no canonical
 235 decomposition into such modes, and choosing the basis would choose the
 236 answer. Applying the scalar threshold to heterogeneous or signed spatial
 237 dynamics requires a justified reduction, and identifying the installed kernel

238 with a cortical field is a separate physical commitment.

239 4 Compatibility: from coordinated activity 240 to a global description

241 4.1 Why common timing is insufficient

242 For overlapping patches U and V , compatibility means that restricting either
243 local description to $U \cap V$ gives the same object. Descriptions can agree on
244 shared quantities while contributing different information elsewhere. Com-
245 patible sections of the specified sheaf have a unique global extension. On
246 a finite cover, local finite spatial measures agreeing as measures on every
247 overlap also have a unique finite-measure extension whose sheaf image is the
248 glued section. No mass-one normalization is imposed.

249 In the cup example, visual and reaching descriptions must express a
250 shared quantity, such as cup location, in common coordinates. Restriction
251 retains that quantity while dropping details particular to each description.
252 Agreement permits surface appearance and hand configuration to contribute
253 to one situation. The mathematical counterexample assigns the same phase
254 to every patch while two local mass profiles disagree on a shared site. Those
255 profiles have no common extension. Because they are chosen independently
256 of phase, synchrony places no constraint on their disagreement.

257 This distinction limits the biological interpretation. Receptive territories

258 and content domains must be related to spatial patches, and the descriptions
 259 must support the required extension. Matching overlap marginals of de-
 260 coded probability laws need not determine a unique global joint distribution
 261 (Abramsky & Brandenburger, 2011).

262 The cover is itself a choice, and the construction constrains it in one way
 263 only: synchronized patches agree where they overlap. Where they do not
 264 overlap there is nothing to compare, so a cover by pairwise disjoint patches
 265 satisfies that obligation for *every* assignment of local descriptions and phases
 266 (`ofDisjointCover`) and the gluing theorem still returns a unique global sec-
 267 tion, which is the tuple it was handed (`glue_unique_of_disjoint`). Member-
 268 ship of the class then reports nothing about the system beyond the patches
 269 covering it. What a gluing claim asks of a cover is therefore a non-degeneracy
 270 the theorem does not assume: that distinct patches meet (`HasNonempty-`
 271 `Overlaps`), and that what they share is a quantity fixed independently of
 272 the agreement it is being used to establish.

273 4.2 When coherence does constrain content

274 A declared relation between phase and content gives a positive result. Let
 275 contents take values in a real normed vector space $\mathcal{C}$. Patch i reads an
 276 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 277 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 278 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 279 sites; their equality on overlaps is a supplied relation between descriptions of

280 the shared quantity. The order-parameter identity $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$
 281 implies $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$ for each pair. Consequently every
 282 overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1 - r^2}. \quad (6)$$

283 Here discrepancy is the norm of the difference, and r and N are read on
 284 whichever population the bound is applied to. That choice is what decides
 285 whether the estimate says anything. A patch carries an order parameter of
 286 its own, and reading the bound there carries that patch's resultant and that
 287 patch's site count (`chord_le_of_patch_coherence`, `compatible_of_patch_`
 288 `coherence`), so a large population observed through small patches is con-
 289 strained by the small count. Read instead on the whole array, the factor N
 290 makes the estimate weak, because it draws a pointwise guarantee from an
 291 average over N^2 pairs. At $r = 1$ the observations agree exactly at either
 292 scale. A shared scalar encoder is a special case. Dropping the overlap rela-
 293 tion permits disagreement even at perfect locking. A common nonexpansive
 294 linear projection preserves agreement, but agreement of a projection does
 295 not establish agreement of the full vector.

296 What Eq. (6) bounds is a chord, which never exceeds 2, so the estimate
 297 constrains anything only where $\sqrt{2} N\sqrt{1 - r^2}$ falls below that. The winding
 298 sweep of Section 8.3 fixes both ends of that range on one sheet. Read on the
 299 sweep's own patch, 625 sites at patch-local order 0.9015, the bound is $382.5 L$

300 and constrains nothing; holding that order fixed, the count alone brings it
 301 below the maximum chord only at 3 sites or fewer. A nearest-neighbour pair
 302 of the same winding, the smallest patch there is, gives $0.1388 L$. Shrinking
 303 patches therefore thins the overlaps at the same rate as it tightens the bound,
 304 and the two ends are not reached by the same cover. At a columnar grain
 305 the estimate constrains agreement inside one locally synchronous population
 306 and is vacuous across the distant territories whose agreement unity is about.

307 That vacuity is the uniform bound's, and three results decline the step
 308 that puts N there: a sum of N^2 nonnegative terms bounded by its total,
 309 which permits all the disorder in a population to sit in the pair the conclu-
 310 sion is about. Asking instead how *many* pairs can be far apart is Markov's
 311 inequality on the same sum, and its answer carries no population count: at
 312 order r the fraction of pairs separated by a chord of c or more is at most
 313 $2(1 - r^2)/c^2$ at any size of population (`chord_fraction_le`). The same sum
 314 read by rows bounds the fraction of *sites* that are far from a fixed fraction
 315 of the population (`site_fraction_le`). Read on the patch above, where the
 316 uniform bound gives $382.5 L$ and constrains nothing, this gives a fraction
 317 at most 0.37 of pairs disagreeing by more than L , and it improves like the
 318 square of what the uniform bound improves like. It is a different quantifier
 319 and not a stronger statement: a count names no pair, the exceptional set
 320 it permits is unlocated and may be exactly the sites a cover's overlaps sit
 321 on, and a guarantee about a named overlap therefore still needs the uniform
 322 bound at its own grain.

323 Distance is reached by chaining rather than by counting. Index the cover
 324 by its nerve, in which two patches are adjacent when they share a site.
 325 The chord is a metric, so a bound β on the separation inside every patch
 326 composes along a walk: a walk of k hops bounds its endpoints by $(k + 1)\beta$
 327 (`chord_le_of_patch_walk_coherence`). The count is $k + 1$ and not k because
 328 both endpoint sites pay a step inside their own patch while the shared sites
 329 are interior to the chain, and the content statement follows for any cover
 330 whose nerve has diameter k (`compatible_of_patch_nerve`). No patch need
 331 contain both sites, which is the situation of every pair of territories far enough
 332 apart, so shrinking patches and spanning territories are two ends of one chain
 333 rather than two ends of one bound. What it costs is that the accumulation is
 334 linear and nothing makes it sublinear: at the sheet's nearest-neighbour value
 335 $\beta = 0.1388 L$ the chain stays inside the maximum chord for 13 hops, which
 336 is 14 patch diameters and 0.219 mm of that sheet. The territories whose
 337 agreement the compatibility claim is about are separated by centimetres,
 338 which is two orders of magnitude beyond where this chain saturates; the route
 339 reaches further only on a cover whose patches are larger or more coherent,
 340 and both are measurements. Connectivity of the nerve is a property of the
 341 cover, supplied rather than derived, and a cover whose nerve is disconnected
 342 supports no conclusion between its components.

343 The third result reads no resultant at all. Where a configuration is
 344 frequency-locked to a common rotation Ω , its phase differences are not free:
 345 each site's detuning is balanced by its coupling term, $\omega_i - \Omega = \sum_j K_{ij} \sin(\theta_i -$

θ_j), which is the Kuramoto equations at a rigidly rotating solution rather
 than a further postulate. Pairing that balance with the configuration itself,
 symmetrizing, and applying Jordan's inequality termwise turns it into the
 coupling graph's Dirichlet form, whose algebraic connectivity λ_2 then bounds
 the spread about the mean phase $\bar{\theta}$ by $2\lambda_2\|\theta - \bar{\theta}\| \leq \pi\|\omega - \Omega\|$ (`spread_le_`
`of_frequency_locked`). Read at a named pair it gives a content residual
 of $L\pi\|\omega - \Omega\|/(\lambda_2\sqrt{2})$ with no population count, no patch and no cover ge-
 ometry in it (`chord_le_of_frequency_locked`, `compatible_of_frequency_`
`locked`), and it improves as the coupling graph becomes better connected,
 which is the dependence a resultant cannot express. Where it is informative
 is fixed by the same hypothesis that licenses it: confinement to a quarter
 turn already bounds every chord by $\sqrt{2}$ with no coupling data at all, so the
 spectral reading adds something exactly where $\lambda_2 > \pi\|\omega - \Omega\|/2$. Four things
 are declared and none is discharged: λ_2 is hardware data in the standing of κ ;
 the locked configuration is assumed, and nothing here proves one exists; the
 quarter-turn confinement is assumed and is not implied by locking; and the
 detuning enters in the population's ℓ^2 norm, so a population whose frequency
 spread grows with its size pays for that growth through the data. What is
 removed is the unconditional factor, not the physics of heterogeneity.

Structural connectivity is a declared graph and is what supplies λ_2 . Mea-
 sured functional connectivity supplies it no better: it is an estimate, so it
 inherits the decoder failure modes of Section 8, a shared prior shrinking esti-
 mates toward agreement and regional bias manufacturing disagreement, and

369 feeding an estimate to a hypothesis strengthens no conclusion drawn from
 370 it. Using it to choose the cover is worse than neutral, being the move of
 371 Section 4 that empties a gluing claim, in different clothes.

372 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 373 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 374 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 375 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

376 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 377 duce an initial mismatch. Approximate observations leave a nondecaying
 378 term in the error bound. Taking a uniform bound on Eq. (6) connects this
 379 update to the phase process.

380 4.3 What approximate agreement determines

381 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
 382 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
 383 Weights that vanish outside their patches and sum to one at each site select
 384 a global profile by weighted averaging. The selection lies within ε of each
 385 patch there, and any two such selections differ by at most ε , independently
 386 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
 387 $L\varepsilon$ across selections.

388 At zero tolerance, the selection is the unique exact extension. At every

389 positive tolerance there are families with no exact extension; selection then
390 gives an approximation whose dependence on weights is bounded. Supple-
391 mental Material gives a three-site witness attaining the bound. The mass-
392 profile construction uses the scalar case of the content dynamics and addi-
393 tionally requires nonnegativity of the initial profiles and observations. The
394 general vector result supplies no identification with the spatial-measure sheaf.

395 These constructions concern declared objects: a specified cover, contents
396 valued in a normed space, and encoders related on overlaps. They select
397 neither the neural cover nor the decoder, and identifying their contents with
398 perceptual content, or establishing that the identification bears on experi-
399 ence, is a further modelling task; Section 8 sets out what a measurement of
400 it would have to establish.

401 **5 Self-representation: a state that models it-** 402 **self**

403 **5.1 The reflexive construction**

404 Compatibility gives a global state. Reflexivity relates it to an encoding within
405 that state. Equip global sections with a complete metric. If the additional
406 restriction-resonance condition identifies encoding with restriction to a region

407 $\mathcal{A}$, the reconstruction map is

$$F(s) = \text{readout}(s|_{\mathcal{A}}). \quad (8)$$

408 The region and readout are fixed across the state space. If F contracts and
 409 fixes the section s_* formed from the local descriptions, Banach’s theorem
 410 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89 sup-
 411 plies the contraction and $F(s_*) = s_*$; restriction resonance is separate. Su-
 412 percritical coupling makes the stipulated contraction factor $\Lambda = e^{-(K-2D)\tau/2}$
 413 less than one for a relaxation time $\tau > 0$. Constructing a map with that
 414 factor remains a modelling obligation.

415 In the cup example, the global state includes the represented scene and the
 416 perceiver’s relation to it. A local encoding must retain enough information
 417 for the readout to recover that modelled state; cup position alone would be
 418 insufficient if hand configuration and viewpoint could not be reconstructed.

419 For the same map with Lipschitz factor $\Lambda < 1$, approximate reconstruc-
 420 tion gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - \Lambda}. \quad (9)$$

421 At the stipulated rate this radius grows as $2\varepsilon/((K-2D)\tau)$ near threshold, for
 422 fixed positive ε and τ . Exact uniqueness survives while the error guarantee
 423 weakens.

424 5.2 Reconstruction across a family of states

425 Sensitivity can be required quantitatively. Declare relevant macrostates, an
 426 encoder E and readout R , and require $d(s, R(E(s))) \leq \varepsilon$ throughout that
 427 family. If two states have the same code, the triangle inequality gives

$$E(s) = E(t) \implies \max\{d(s, R(E(s))), d(t, R(E(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

428 Thus M relevant states pairwise more than 2ε apart require at least M
 429 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
 430 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
 431 existing self-prediction map.

432 This is an additional accuracy criterion on a family, with no requirement
 433 for multiple exact fixed points. A blind encoder or constant readout fails it
 434 for separated states despite having a unique fixed point.

435 Where the encoding is read as the state's own restriction to a region, the
 436 criterion becomes a condition on the region. Restriction resonance requires
 437 the encoder and that restriction to identify the same pairs of states, so a
 438 boundary reconstructing two relevant states more than 2ε apart while its re-
 439 gion reads them identically is not resonant (`not_isRestrictionResonance_`
 440 `of_reconstructs`). Read forwards, a resonant boundary that reconstructs
 441 a pairwise separated family restricts injectively on it (`restrictToAvatar_`
 442 `injOn_of_isRestrictionResonance`). Which states a region reads identi-
 443 cally is fixed by where the region sits, before an encoding is chosen, so the

444 resolution the family demands is the region's and no readout supplies it. The
 445 implication runs one way: a region that does resolve the family is not thereby
 446 shown to carry a resonant encoding, and E89 supplies no such resolution.

447 Those two demands are jointly satisfiable with nothing folded anywhere.
 448 Take the region to be the whole substrate: restriction is then the identity, so
 449 the boundary is resonant, reconstructs every declarable family exactly, and
 450 restricts injectively on all of them (`topAvatar_satisfies_obligations`). A
 451 properness condition on the region (`IsProperAvatar`) is therefore a separate
 452 requirement rather than a consequence of resonance and resolution, and what
 453 a fold into a local region is asked for. Its quantitative form replaces the
 454 alphabet by a metric. With an L -Lipschitz readout, a family reconstructed
 455 within ε whose members lie pairwise more than $d_{\min}$ apart has δ -separated
 456 codes whenever $\delta L \leq d_{\min} - 2\varepsilon$, so the family is no larger than the δ -packing
 457 number of what the region carries (`encard_le_packingNumber_restrict-`
 458 `ToAvatar`). That is the shape a continuous family of states needs, and at
 459 the whole substrate it says nothing, because the region's range is everything.
 460 The gain L is declared and derived here for no physical readout, and at
 461 unbounded gain the bound says nothing.

462 **5.3 Agreement after a change takes communication**

463 Snapshot compatibility does not specify how quickly descriptions respond
 464 when a shared quantity changes. In a synchronous network whose updates
 465 read only the site's own state and declared neighbours' messages, two initial

466 configurations agreeing on a site’s causal past through T rounds produce the
467 same state there at round T . The deadline consequence is the one Section 2
468 states, and it follows from the reconstruction bound in Section 5.2: a report
469 from that site is a readout of a code that the two interventions leave identical.

470 A three-site delay line witnesses failure after one round and exact report-
471 ing after two; cutting the path prevents delivery at every round. A report
472 may still happen to be correct about one fixed world. The bound concerns
473 responsiveness across interventions.

474 Throughout this section *prediction* names a reconstruction map, and the
475 relevant family of states, the metric, the tolerance, the communication graph
476 and the deadline are all declared inputs, as are the represented variables and
477 the fidelity counted as accurate. Reading the map as temporal neural pre-
478 diction would require dynamics; reading the resulting code count as physical
479 memory capacity would require a hardware model, which a state-dependent
480 readout would have to extend.

481 **6 What the conditions constrain in a digital** 482 **candidate**

483 The results of Section 2 hold for any candidate, and for a large language
484 model (LLM) running on graphics processing units (GPUs) four of them are
485 checkable rather than merely stateable. Those four are where this framework
486 bears on claims made for such systems. It does not exclude them, and no re-

487 sult here could; what it does is fix what a claim of unity or self-representation
488 commits its maker to measuring.

489 **Codes, not reports.** Reconstruction across a family is a counting con-
490 straint before it is anything else. If two macrostates the candidate is said
491 to represent satisfy $E(s) = E(t)$ while $d(s, t) > 2\varepsilon$, Eq. (10) rules out re-
492 constructing both within ε , whatever the readout. A system’s reports are
493 readouts. Identical verbal reports across two conditions are therefore consis-
494 tent with identical internal codes and settle nothing about the states. The
495 converse claim—that the system represents its own condition across a de-
496 clared family—asserts that its internal code separates that family, which is a
497 statement about activations under intervention and not about text. Naming
498 the channel makes the count concrete. Where the code is the token sequence
499 a model emits—which is what its next step reads—the alphabet is the de-
500 ployed vocabulary (`tokenChannel`), and a family reconstructed through it
501 is no larger than the vocabulary size raised to the number of steps (`card_`
502 `le_card_tokens`). Over an extended exchange that number is large and the
503 bound slack. At a single step it is one vocabulary’s worth, so claiming that a
504 single emission reports the model’s own condition across a declared family as-
505 serts that the vocabulary separates it (`card_le_card_tokens_one`). A claim
506 resting on activations rather than on text has declared a different channel,
507 to which the same count applies with that channel’s alphabet. The measure-
508 ment follows from the statement: declare the family and the fidelity first,

509 then test whether the encoding region distinguishes its members.

510 **The region, not the readout.** Where the encoding is read as the state's
511 restriction to a region of the machine, the demand falls on the region before
512 any readout exists. For a decoder-only transformer the causal mask makes
513 the network an instance of the update model of Section 5: the activation
514 at position t and layer ℓ reads positions at most t at layer $\ell - 1$ and nothing else. A region's causal past is therefore fixed by the architecture rather
515 than chosen by interpretation. That instance is formal, and sharper than
516 slow growth: the causal past of position t stays inside the positions at most
517 t at every depth (`mask_ball_subset_1e`). The deadline bound of Section 5
518 therefore fires at every deadline at once rather than at a chosen one, its communication graph given rather than assumed (`mask_no_guarantee`). That is
519 a statement about one forward pass; across steps a model does read its own
520 prior output, which is the channel the preceding paragraph counts. Read
521 against the coupling proposed here the asymmetry runs both ways, and neither way is an advantage. A kernel of full support has no neighbourhood
522 structure for the bound to bite on: after one round the causal past is the
523 whole substrate, so the hypothesis that fires the obstruction is unsatisfiable
524 at any deadline but zero (`not_outside_past_of_isFullSupport`). A physical
525 field propagates at a finite speed, so treating one that way is a calibration
526 against the phase time scale and not a theorem. What the comparison does
527 fix is where the bound has force. Not on the attention graph, where a posi-

531 tion reads every position at or before it and the fan-in is the context length.
 532 On the physical interconnect, fan-in is genuinely bounded: a graph whose
 533 sites each hear from at most ν others reaches at most $(T + 1)\nu^T$ sites in
 534 T rounds (`card_ball_le_mul_pow`), so a model sharded across many devices
 535 pays a number of rounds growing like $\log_\nu$ of the device count. Buying that
 536 deadline back costs wiring: reaching every site in one round requires a device
 537 hearing from all but one of the others (`le_degree_of_reaches_one`). Both
 538 ν and the device count are declared, as the communication graph and the
 539 deadline already are, and what the count bounds is the graph a guarantee
 540 requires rather than what hardware is buildable. A claim that a model re-
 541 ports on its own processing consequently carries an obligation to say which
 542 positions and layers the report’s causal past contains. A report that appears
 543 to track an intervention outside it indicates a confound or a different region,
 544 not a faster channel.

545 **Agreement about what.** Unity requires the candidate’s local descriptions
 546 to agree on independently identified shared quantities (Section 4). Common
 547 timing, a shared residual stream and fluent output supply none of that, for
 548 the reason Section 2 gives: coordination bounds content disagreement only
 549 through encoders already related on overlaps. Which quantities two parts
 550 of a model describe in common is a decoding question, carrying the failure
 551 modes of Section 8. The cover is a choice, and choosing it to secure agreement
 552 empties the claim.

553 **What is not excluded.** Digital representations can satisfy compatibility,
 554 so finite hardware is compatible with the gluing construction and none of
 555 the above is a no-go for machines. The coupling-gated route E56–E78 is a
 556 different matter: it requires a calibrated physical coupling, its noise regime
 557 and a connection from coordination to content agreement, and an attention
 558 matrix or a shared clock supplies none of these identifications. Computing a
 559 field model likewise leaves its realization in the computing substrate to be es-
 560 tablished. Those physical obligations fall on the cortical proposal identically.
 561 The capacity comparison those support comparisons lack is available, and it
 562 favours neither side by construction. A continuum code space read at reso-
 563 lution δ holds at least its volume divided by the volume of a resolution cell,
 564 so it out-resolves an alphabet of 2^b codes exactly when that volume exceeds
 565 2^b cells (`two_pow_lt_packingNumber_of_lt_measure`). Below that threshold
 566 the inequality runs the other way, and at a coarse enough resolution the re-
 567 gion holds a single code. What decides the comparison is the volume, the
 568 resolution and the dimension, all measured quantities, and continuity decides
 569 nothing on its own: a continuum code space read at a finite resolution is a fi-
 570 nite alphabet whose capacity is a bit count. For the same reason this account
 571 credits an analog medium with no computation beyond a Turing machine’s.
 572 Real-valued states carry unbounded capacity only at unbounded precision,
 573 and the noise floor $D > 0$ that Eq. (1) is stated against removes it.

574 The installed-energy condition of Section 3.2 is restricted in the same
 575 way. It constrains a macroscopic candidate only through κ , which is hard-

ware data supplied by mode profiles and prices, and Supplemental Material reads the condition against a cortical metabolic budget to return the conversion factor it would require rather than a verdict. The hardware comparisons there have similarly restricted scope: fixed wiring can impose a correlation disadvantage, and point-supported kernels vanish under an atomless continuum measure, but neither comparison makes the representation conditions unattainable on GPUs. A universal exclusion would need a necessary condition for consciousness and a proof that every relevant implementation violates it, and the present composition supplies neither.

Whether experience requires a living substrate is the explicit subject of biological naturalism (Seth, 2026a,b). On that view the properties grounding experience are those of self-sustaining organisms, and simulating them does not secure them. The conditions stated here name no biological mechanism, but computation described as such does not satisfy them either: a candidate must exhibit a physically realized coupling in a measured noise regime, agreement about independently identified shared quantities, and an internal reconstruction meeting a declared fidelity. Whether a realization of those conditions that is not alive would be accompanied by experience is the point at issue, and these results do not settle it. A related position locates the difference in the mode of computation rather than in life as such: biological systems compute across scales that do not separate, and alongside discrete operations carry continuous-valued dynamics belonging to their physical medium (Milinkovic & Aru, 2026). The coupling condition stated

599 here is one measurable instance of that requirement, fixed to a physical field
600 in a declared noise regime, and a description of such a field does not satisfy
601 it.

602 The neuroscientific case against *human-like* consciousness in text-only
603 LLMs concerns embodied sensory interaction, thalamocortical organization
604 and biological regulation linking perception and action to survival (Aru et al.,
605 2023). Their absence supports skepticism, conditional on their causal rele-
606 vance to experience. Those mechanisms are not requirements of the abstract
607 gluing and reconstruction definitions, and workspace-like representations re-
608 ported in some LLMs (Gurnee et al., 2026) supply candidate organization to
609 investigate. What this framework adds to that investigation is an order for
610 the questions: declare the family of states and the fidelity, fix the encoding
611 region and read its causal past off the architecture, and only then ask what
612 the system reconstructs. A claim that survives those three steps is a claim
613 about the machine; one that does not survive them was a claim about its
614 output.

615 **7 Thermodynamics and composition**

616 The phase and representation results concern coupling, content disagreement
617 and reconstruction error. Thermodynamics constrains the physical opera-
618 tions that could implement them, and it does so from the side rather than
619 along the spine: no edge below derives coherence, compatibility or reconstruc-

tion from a heat bound, and the resource condition these edges terminate in
is necessary without being restrictive for any macroscopic candidate (Sec-
tion 3.2). What this half contributes is a ledger in which a resource claim
is stated in checkable units. Finite-register capacity obeys $H(p) \leq \log |X|$;
erasure heat depends on specified system–bath assumptions (Landauer, 1961;
Norton, 2005; Shenker, 2000). A separate boundary result concerns a field
reaching disconnected vacuum components (Kibble, 1976). The composition
connects these objects through the explicit hypotheses in Table 1.

7.1 Feedback changes the information constraint

For passive memory X_t driven by an environmental signal S_t , the predictive
bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (11)$$

The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still
et al., 2012). Zero retained information also satisfies it, so a task and a
learning rule require additional specification. Relating this bound to Lan-
dauer heat requires a common thermal scale and budget, and memory waste
bounded by erased entropy.

An acting agent changes the signal it will receive. Specify policy, world,

637 observation and memory-update channels. For their finite laws,

$$\begin{aligned}
I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\
&\quad - I(X_t; S_{t+1} \mid S_t).
\end{aligned}
\tag{12}$$

638 The second conditional term vanishes in the passive model. With feedback,
639 the signed difference can be negative as action creates correlations with the
640 future. A causal reading requires an environmental state sufficient to describe
641 the relevant dynamics.

642 For a controlled substep, the active thermodynamic bound instead uses
643 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F \| P_R) =$
644 $\Delta H(X, S) + \langle Q \rangle / (k_B T) \geq 0$ under local detailed balance, with thermal scale
645 $k_B T > 0$ and heat positive into the reservoir (Horowitz and Esposito, 2014;
646 Ito and Sagawa, 2013). Every forward path of positive probability must have
647 a positive-probability reverse for this finite identity. A missing reverse gives
648 infinite divergence. The supplement develops the support conditions and
649 composes actuation with memory updates on their shared intermediate law.

650 **7.2 Learning and repeated operation need explicit re-** 651 **sources**

652 The finite examples distinguish selecting a feasible policy, relaxing a register
653 toward an encoded reward, and learning an unknown task from observa-
654 tions. The observational learner acquires correlation with an environmental

parameter while its action marginal stays uniform. A continuing model executes actuation, observation-driven learning and reset in sequence, preserving above-chance performance at a positive work cost per cycle.

A phase-derived sensor makes the informational distinction concrete: an informative readout and a blind control have the same phase order but different learning performance. Attaching the learner’s register to the actuator of Section 3.2 ties reward, installed kernel and work to its own executed law. The learning witness rises above the necessary installed-energy bound while its matched control remains on the exclusion line. These finite constructions, detailed in Supplemental Material, leave the evolving feedback between phase dynamics, installation and sensing to be specified.

For that model, common energy observable $\mathcal{E}$, thermal scale $k_B T$, local detailed balance and positive support at each stage give

$$k_B T [H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle \mathcal{E} \rangle_0 - \langle \mathcal{E} \rangle_N \leq B + \langle \mathcal{E} \rangle_0 - \langle \mathcal{E} \rangle_N, \quad (13)$$

where expected cumulative work stays within the supplied allowance B at every prefix. This bounds expectations. Separate constructions track a store on each trajectory, include finite sources, charge sensor clearing and prepend preparation. They specify when a mean allowance represents an available resource; an expected budget alone permits overdrawn paths.

These constructions establish costs for their declared processes. A finite source with a positive net cost permits only finitely many funded cycles,

while continuing operation requires replenishment. Register–bath witnesses also distinguish actual bath-energy transfer from transition-log heat. Identifying a work store, an erasure budget and the installation energy of Section 3.2 requires physical relations that these separate models do not supply automatically.

7.3 What the composition establishes

The passive theorem `chain` and feedback theorem `chain_active` each accept eight connecting implications (Table 1). In the active branch, a named substep at the register’s thermal scale satisfies

$$k_B T > 0, \quad k_B T [H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (14)$$

with B the register’s heat. The upper allocation follows from a ledger whose operations deliver that total heat to one reservoir and whose other operations do not increase joint entropy. Those ledger assumptions are physical inputs. Neither Landauer’s lower bound nor microscopic reversibility supplies them. The regime in which those premises are exhibited is a near-equilibrium one: strictly positive forward and reverse elementary paths, local detailed balance at one thermal scale, and entropy differences and heats of order $k_B T$, the witnesses being two-bit systems with interaction energies of that size. Cortex is not in it. The installed energy of Section 3.2 is 2.4×10^{14} times kT at body temperature over one region and one phase-diffusion time, and a budget B

694 of that size stands against an entropy term of order $k_B T$, so Eq. (14) is
 695 satisfied by any macroscopic candidate and excludes none of them. Reading
 696 the register instead at the scale where its premises are physical, such as a
 697 single channel, relocates the difficulty to E45Active, which would then have to
 698 carry one molecular step’s budget to the convergence of a continuum kernel.
 699 The active branch is an alternative route to the same node rather than a
 700 stronger one, and no result of Section 3.2 passes through it: the installed-
 701 energy condition is derived from E56 and E67.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12 <i>capacity and boundary</i>	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.
E23 <i>boundary to register</i>	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34 / E34Active <i>register to dissipation</i>	Bridge assumption	Passive: predictive memory at the register’s thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations’ compressiveness. Both are exhibited at thermal-scale heats on finite state spaces with positive reverse paths, which is not the regime a cortical register occupies.

Table 1 (continued)

Connection	Status	Required assumption
E45 / E45Active <i>couplings to continuum</i>	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Convergence follows from mesh refinement rather than from the budget, which selects no regime at macroscopic scale. Refinement and physical identifications remain inputs.
E56 <i>field identification</i>	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67 <i>above threshold</i>	Physical commitment	That field operates above the synchronization threshold.
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours. Where the cover's nerve is connected the same edge carries through the sites its patches share, at a residual linear in that nerve's diameter; where the configuration is frequency-locked it carries through the coupling graph's algebraic connectivity λ_2 instead of through any patch count, which asks the cover for nothing and is declared hardware data in the standing of κ .

Table 1 (continued)

Connection	Status	Required assumption
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $\Lambda < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 3.2’s occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

Both branches also take a thermodynamic cover carrying exact overlap compatibility and a reflexive boundary specifying encoding and readout. They conclude **UnifiedSelf**: the glued section is the unique fixed point of self-prediction. Compatibility and accuracy at that section enter as premises; uniqueness follows. Identifying encoding with restriction needs the separate resonance condition of Eq. (8).

Each of the eight edges is supplied rather than derived from the one above it. E12 is equivalent to its conclusion, so capacity and boundary are independent roots rather than a step from one to the other; both E45 discharges take their convergence from refinement data and not from the thermodynamic bound; E78 states that coherence relaxes the supplied cover, and the coherence-to-content estimate of Eq. (6) bounds disagreement only for encoders already shared on overlaps. The inferential content of the composition is its last step, where contraction and a fixed glued section give uniqueness. What composing them establishes is that the premises are consistent, named

717 in one place and machine-checked together, and not that coherence follows
718 from installed energy or self-representation from coherence.

719 Joint witnesses combine a register, boundary, refining mesh and small
720 spatial cover; active witnesses add controlled thermal updates and a priced
721 installation. They establish satisfiability under their stated identifications.
722 Mesh convergence follows from refinement data, and the witnesses establish
723 no common cortical mechanism. Supplemental Material maps the claims to
724 their Lean declarations and records the counterexamples. The development
725 declares no additional axioms beyond its libraries' foundational axioms.

726 8 What the observations establish

727 The composition holds under its premises. Whether observation can estab-
728 lish those premises is a separate question, and the three requirements it
729 connects concern different objects: phase dynamics, local descriptions and
730 internal encodings. Their observables must retain these distinctions. The
731 following controls show which mechanisms remain indistinguishable, which
732 temporal approximations fail, which states one observable cannot separate,
733 and what independent content measurements must establish. Each is a finite
734 numerical regime on a declared model, with full parameters and controls in
735 Supplemental Material, and none of them calibrates a cortical quantity.

736 8.1 Stationary distributions leave the mechanism un- 737 determined

738 Equation (2) gives a concentration–coherence relation with no fitted shape
739 parameter. Estimate r from circular resultant length and a from density
740 shape: inverting R from the measured r would make agreement automatic.
741 Even independently estimated agreement identifies a statistical family shared
742 by different mechanisms.

743 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
744 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
745 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
746 oscillators with prescribed drift $-h(t) \sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,
747 and diffusion D_0 have that same density. Their stationary distributions co-
748 incide window by window despite different causes.

749 Resolved dynamics supply additional information. In the ideal model with
750 continuous paths, quadratic variation accumulates at rate $2D$ and conditional
751 increment means constrain drift. Separating endogenous coupling from a
752 matched common drive additionally requires a measured drive or controlled
753 perturbation; identifying electromagnetic coupling requires evidence specific
754 to that interaction. Supplemental Material gives the equivalence construction
755 and its numerical check.

756 8.2 Finite ramps change the observable onset

757 The square-root law in Eq. (3) describes stationary solutions. A changing
758 coupling requires time for the phase distribution to approach those solutions.
759 In a deterministic mean-field control that crosses the threshold symmetrically,
760 none of the scanned rates meets the declared tracking tolerance: slowing
761 the crossing does not remove the terminal residual under that initialization.
762 A slow control variable therefore supplies no general guarantee of branch
763 tracking at onset. Supplemental Material gives the residual, rate range and
764 admissible rates away from threshold.

765 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas
766 at 8 coupling speeds from $v = 0.1$ to $v = 0.0001$. Escape is the first of three
767 consecutive saved samples at $r \geq 0.2$ after threshold. Faster ramps delay
768 escape (Fig. 3); only 5/32 replicas escape within the fastest window. The
769 6 uncensored legs supply the delay scaling below; the decade-spaced ones
770 among them have mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted
771 onset exponents differ from those of the exact stationary branch evaluated
772 on the same grids. At the fastest speed the ensemble mean never reaches the
773 fit window, so its fitted slope does not estimate stationary onset.

774 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)
775 makes the apparent exponent depend on ramp speed and fitting window.
776 The specified deviation-based critical ramp speed is not bracketed by this
777 sweep. With fluctuating coupling, the relevant stationary comparison also
778 depends on rate: the mean coupling approximates order in the tested fast

779 regime, while averaging stationary order over coupling works for the tested
 780 slow drives confined to one side of threshold. Slow crossing drives can satisfy
 781 neither approximation.

782 **The prediction.** The delay scaling is what this section supports as a pre-
 783 diction, and it needs no absolute calibration. Over the 6 uncensored legs the
 784 mean delay follows $\Delta K \propto v^{0.444}$, with a 95% slope interval of $[0.394, 0.494]$,
 785 which does not contain $1/2$. What that shortfall measures is the escape cri-
 786 terion and not the scaling. Applied to a deterministic mean-field run seeded
 787 at the fluctuation floor, the same estimator returns 0.425 over the same legs
 788 at the same escape criterion, inside the interval on the measurement. That
 789 run has no noise to delay it, so the shortfall is the criterion's: it crosses the
 790 saturation between the floor it starts from and the level it is stopped at, and
 791 tightening the criterion to $r \geq 0.05$ returns 0.490. The bias is reproduced
 792 where the answer is known, which is what licenses reading the measurement
 793 as the law rather than as a competing exponent.

794 Stated as a law rather than as a number: onset delay scales as the square
 795 root of the rate at which coupling crosses threshold. Testing it requires
 796 only that $\dot{K}$ be monotonic in a manipulated rate—not its value, nor the
 797 installation factor κ , nor the field-to-coupling conversion rate γ of Eq. (15).
 798 It is therefore a within-subject design varying the rate of emergence from
 799 anaesthesia alone, each participant serving as their own control. Such a
 800 design must avoid the hazard this sweep exhibits: a fixed absolute escape

801 criterion meets a size-dependent critical fluctuation floor, so the criterion
 802 has to be referred to each recording's own pre-threshold fluctuation scale
 803 rather than to a shared number. The diffusion D is the remaining input,
 804 and Section 8.1 notes that it is estimable from resolved phase traces through
 805 the rate $2D$ at which quadratic variation accumulates, which is a different
 806 quantity from the concentration the propofol exercise could not range over.

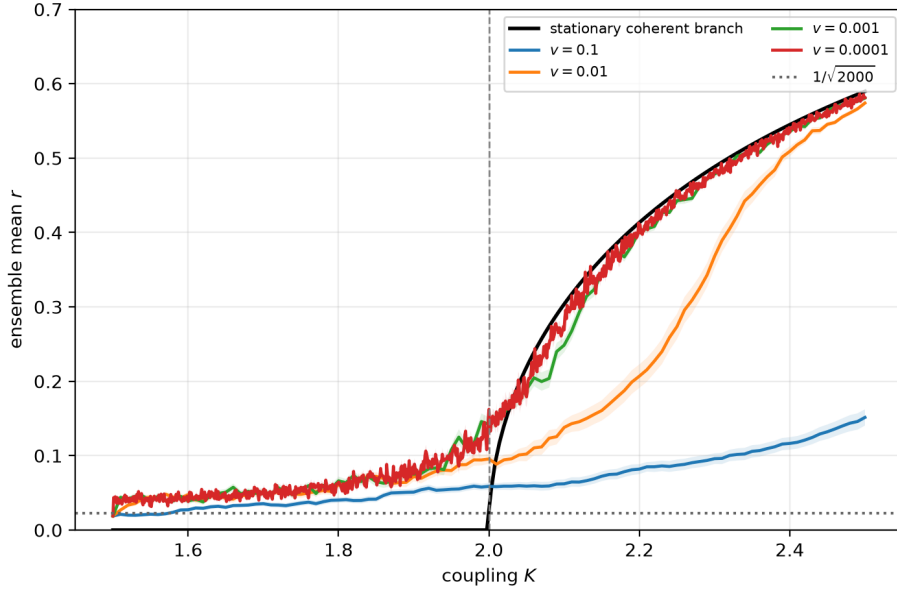


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

807 Additional controls address integration bias, population size and spatial
 808 or signed coupling; the refined signed-network crossing bracket $K_{\text{eff}}/D \in$
 809 $[1.958, 2.009]$ contains 2.

8.3 A winding state leaves the order parameter ambiguous

The order parameter r is indifferent to where a phase sits in space. A field winding once across a periodic sheet therefore has $r = 0$ while every patch of it stays locked. Such a field is not an artificial construction here. On a torus a winding is stationary for any isotropic symmetric kernel at identical frequencies: the coupling term is odd in the separation against a kernel even in it, so the drift cancels pair by pair. `char_is_kuramoto_trajectory` proves this for any finite periodic array, this sheet included. A winding satisfies the positivity and identical-frequency hypotheses. It violates only the quarter-turn initial spread that the convergence result additionally requires before placing the limit at the uniform state.

On the sheet of Section 3's spatial control, a winding survives below an interpolated decay length of 0.1871 mm and unwinds above it (Fig. 4). That boundary is interior to the 0.1–0.3 mm band. At 0.1 mm the same sheet at the same coupling, diffusion and kernel reaches $r = 0.9657$ from a uniform start and $r = 0.0083$ from a winding one. Both are stationary, with patch-local order 0.9015 and no winding defect. At fixed K the observable therefore takes two values spanning nearly its whole range. Coupling does not determine the resultant, and a low resultant does not by itself distinguish weak coupling from a winding field.

Under the quenched frequency spread of 1.5 rad s^{-1} that the spatial con-

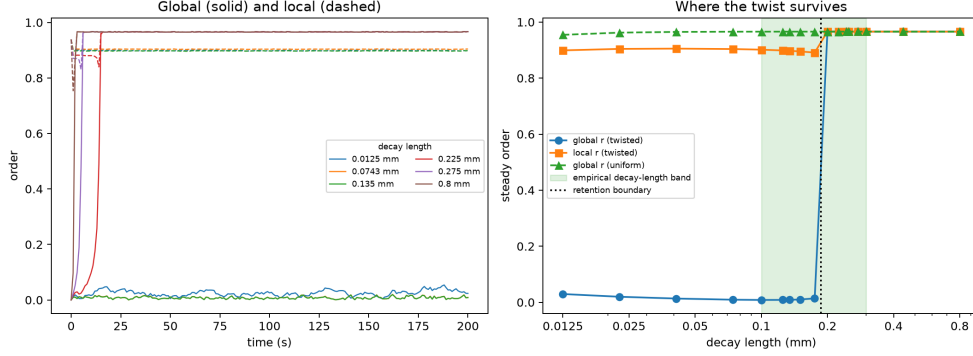


Figure 4: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

832 trol carries, the boundary moves to 0.1421 mm. The same comparison at
 833 0.1 mm gives 0.9418 against 0.0348, holding across 3 frozen realizations.
 834 Heterogeneity shortens the range that sustains a wave without removing it.

835 Which windings persist has a criterion behind it. Linearizing the phase
 836 model about a winding returns a coupling of the same translation-invariant
 837 form, each separation reweighted by the cosine of the phase the winding
 838 advances across it (`hasDerivAt_charJacobian`). That reweighted kernel is
 839 again even in the separation, so the same characters diagonalize it and ev-
 840 ery mode’s rate is one of its eigenvalues. Where the winding advances by
 841 at most a quarter turn across every separation the coupling reaches, the

842 reweighted kernel is nonnegative and no rate is positive (`charJacobian_`
 843 `stable_of_quarter_turn`); where some rate is positive, that mode grows
 844 without bound (`charJacobian_growing_mode`). Moving coupling mass to
 845 longer separations is what carries a phase advance past a quarter turn, and
 846 that is the mechanism the decay-length sweep varies. The criterion bounds
 847 rates rather than trajectories: asymptotic stability is not proved, so the
 848 states that persist occupy a narrower range than it admits, and the bound-
 849 ary quoted above is measured and not derived from it. The same kernel is the
 850 potential's curvature, so a winding at $r = 0$ is not the global minimum the
 851 convergence result places its limit at, and one carrying a growing mode is not
 852 a local minimum either: that result's quarter-turn hypothesis must exclude
 853 a winding rather than merely decline to cover it (`char_not_potential_min`).

854 An amplitude layer decides which of these states exist, and leaves what
 855 the resultant reports unchanged. Every oscillator of the phase model sits on
 856 the unit circle, so a field with an amplitude node has no representation in
 857 those dynamics. The real Stuart–Landau extension supplies one on the same
 858 coupling, and leaves intact the gradient structure the convergence results rest
 859 on. The state becomes $z_i \in \mathbb{C}$ with $\dot{z}_i = (\Gamma - |z_i|^2)z_i + \sum_j A_{ij}(z_j - z_i)$. Its
 860 phase equation is the Kuramoto equation with each coupling weighted by a
 861 ratio of amplitudes (`amplitude_phase_is_kuramoto`). The phase model is
 862 this one with the amplitude ratios set to one, rather than a separate object.

863 What the added degree of freedom decides is which windings exist at all.
 864 On a periodic array of any dimension, this sheet included, a winding requires

865 $\Gamma + \lambda > 0$, with Γ the layer's linear gain. Here $\lambda = \sum_d f_d(\cos \Delta\theta_d - 1)$ is
 866 the eigenvalue the kernel supplies, summed over separations, with $\Delta\theta_d$ the
 867 phase the winding advances across d (`exists_charState_iff`; on the ring
 868 $\Delta\theta_d = 2\pi qd/n$). For the nonnegative kernels used here $\lambda \leq 0$, and moving
 869 coupling mass to longer separations lowers it further. A broad enough kernel
 870 therefore removes the winding outright. The sweep above reaches that same
 871 boundary by a different route: the phase model retains every winding at
 872 every coupling, and loses this one under integration instead. The two layers
 873 bound the same regime from opposite sides. A phase recording does not
 874 separate a state the amplitude layer excludes from one the dynamics fails to
 875 sustain.

876 Three limits bound the layer for a reader of the cortical literature. These
 877 states are windings and not spirals, and on the sheet that is a theorem rather
 878 than a shortage of dimensions. Equal modulus at every site puts the phase
 879 circulation at exactly zero on every plaquette, and on a periodic sheet the
 880 circulations of *any* field sum to zero, so a lone defect is not a state of this
 881 geometry at all. Relaxational dynamics has no propagation speed, which is
 882 the quantity a cortical wave measurement leads with. And $|z|$ is a normal-
 883 form radius, which a measured envelope is not; this work identifies the two
 884 nowhere. Supplemental Material gives the sweep, its uniform control, the
 885 amplitude layer, the standing wave it admits inside the band, the potential's
 886 curvature at a winding and the defect theorems, together with the remaining
 887 limits of both.

888 A vanishing resultant is the absence of global agreement and not of agree-
 889 ment. Eq. (6) read on a nearest-neighbour patch of a winding gives a discrep-
 890 ancy bound of $2\sqrt{2}L|\sin \Delta\theta|$, with $\Delta\theta$ the phase advanced across the patch,
 891 which falls to zero as the same winding is spread over more sites (`chord_le-`
 892 `of_char_patch`), while the global form of the same bound is vacuous at $r = 0$.
 893 Encoders shared on overlaps therefore still bound content disagreement in-
 894 side a patch of a state that no global observable calls coherent. The winding
 895 number itself stays unconnected to content. It is, however, a property of the
 896 state rather than of the cover read on it. On a ring the whole turns dropped
 897 by the winding's affine local lifts are zero at every site but the wrap, where
 898 the residue resets, and they sum around the loop to q ; a nonzero q obstructs
 899 every global real-valued phase lift, so one state at one coupling has resultant
 900 exactly zero and degree q (`winding_degree_obstructs`). Reweighting the
 901 sites moves the first quantity and redrawing the loop's intermediate transi-
 902 tions leaves the second where it was. The scope is the ring's: this is a loop's
 903 winding and not a spiral, the transitions are those of the affine lifts rather
 904 than of an unwrapped measurement, and the integer is identified with no
 905 cortical observable.

906 This qualifies what a phase trace can establish. An onset criterion stated
 907 in r alone registers a change of basin as readily as a change of coupling.
 908 The protocol of Section 9.3 therefore needs a spatial observable beside the
 909 resultant.

910 8.4 Coherent adaptation can lose environmental align- 911 ment

912 An adaptive coupling rule might supply representational structure, but its
913 objective must reward the relevant agreement. For symmetric coupling, pair-
914 wise sine terms cancel in the population mean drift. Minimizing squared drift
915 at that fixed mean therefore rewards equalizing drift across nodes. In the
916 frequency-cluster control, this favors coupling between unlike-frequency clus-
917 ters, whereas the diagnostic environmental template rewards coupling within
918 clusters.

919 The formalization proves descent of the squared-drift objective for a spec-
920 ified gradient flow at fixed phases. Its functional differs from thermodynamic
921 entropy production: at a coherent stationary density the probability current
922 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
923 rial gives the operator, its gradient and the conditions of the descent result.

924 The finite control co-evolves noisy phases and symmetric, nonnegative
925 couplings at fixed total resource. Its template is a diagnostic covariance of
926 latent signals, never a training input. Across three seeds, second-half order
927 remains between 0.9167 and 0.9222, while the within-cluster to between-
928 cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 5).
929 Random-update, permuted-gradient and frozen-kernel controls are reported
930 in Supplemental Material.

931 The immediate post-update objective reduction disappears when aver-

932 aged over complete update intervals. A prespecified grid of 6 rate–cadence
933 combinations contains no case meeting the 5% sustained-reduction criterion
934 while retaining coherence and losing alignment. The finding is coherent adap-
935 tation away from the specified structure, with no demonstrated sustained cost
936 reduction.

937 Kernel alignment and overlap compatibility are different observables.
938 This control addresses acquisition of its designated template under its par-
939 ticular objective. Learning models can explicitly include data in an objective
940 (Corberi et al., 2026); an account of agreement about the cup must specify
941 the content its learning rule is meant to preserve.

942 **8.5 Content agreement needs independent decoders**

943 A neural test needs independently identified domains, local descriptions and
944 restriction maps. Two decoders should estimate a shared content on over-
945 lapping domains before their agreement is examined. Discrepancy requires
946 a null under decoding noise and controls using deliberately compatible and
947 incompatible examples. An implementation on constructed data separates
948 these cases at $AUC = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by
949 trial; the compatible construction scores 0.476 on an interval containing one
950 half. Shared-prior shrinkage can nevertheless manufacture agreement, and
951 regional bias can manufacture disagreement. Per-patch decoding diagnostics
952 detect these failures; the compatibility statistic alone does not.

953 Phase agreement cannot replace that measurement. On constructed con-

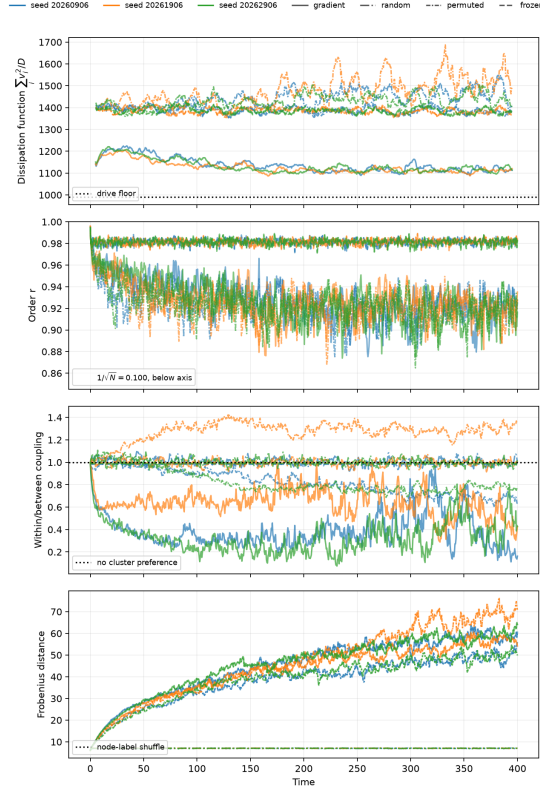


Figure 5: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

954 tent, phase carries a share 1.000 of the content entropy when content is built
955 from it and 0.000 when it is not. A phase-derived statistic reports com-
956 patibility on phase-locked patches whose local descriptions disagree. The
957 implemented content statistic detects their incompatibility. These controls
958 validate a measurement on known constructions; a neural content model and
959 decoder remain to be established.

960 Choosing the cover also chooses a candidate subject: trivial descriptions
961 can agree while carrying little content, and a partition can conceal disagree-
962 ment. The useful choice must therefore be constrained by what neural ac-
963 tivity encodes. The exact composition assumes compatibility of its supplied
964 cover. The content mechanisms above identify additional hypotheses under
965 which agreement can arise; they leave the empirical choice of content and
966 the connection to that cover explicit.

967 **8.6 What an informative phase-density measurement** 968 **requires**

969 The concentration range and the dependence structure decide whether the
970 concentration–coherence relation can be tested at all. Under the specified
971 log-density estimator and bias-floor criterion, discrimination at the observed
972 concentration ceiling $a = 0.542$ requires 3000 independent sites in the scanned
973 designs, rising to 31000 when sites share a mean in groups of 10. These
974 counts describe that estimator and dependence model; the intended recording

975 requires its own calibration. Supplemental Material reports the bin-count
976 controls and the full dependence-aware design.

977 That ceiling comes from exercising the estimators on public propofol-
978 sedation EEG (Bajwa et al., 2025), pooling channels in 100-ms windows.
979 Bipolar and common-reference montages produce different residuals. The
980 bipolar results are compatible with the Bessel relation over a concentration
981 range too narrow to distinguish it from its linear approximation at the ob-
982 served scatter, which is the range the design scan takes as given. Temporal
983 pooling changes the estimand, and sedation recordings leave the awakening
984 hypothesis untested.

985 An informative dataset therefore needs sufficient concentration range or
986 precision, calibrated estimators and an observation model for dependence,
987 source mixing and reference choice. Public mouse calcium imaging provides
988 thousands of spatially located neurons across waking, sleep and anesthe-
989 sia (Oomoto et al., 2026), but its 7.65 Hz frame rate is below the phase-
990 estimation band used here. Site count, state coverage and bandwidth are
991 separate requirements.

992 **9 The cortical hypothesis and an awakening** 993 **test**

994 The cortical proposal identifies a physical source of coupling and asks whether
995 its changes help explain recovery. The tests in Section 8 determine what

996 observations must distinguish before a phase trace can support this identifi-
997 cation.

998 **9.1 Physical identification and calibration**

999 The cortical hypothesis identifies part of effective coupling with endogenous
1000 extracellular electromagnetic interactions. Fields can modulate spike timing
1001 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
1002 propose coordination of distributed neural information (Pinotsis & Miller,
1003 2022; McFadden, 2020). Phase organization under propofol motivates study-
1004 ing this regime (Bastos et al., 2021), but does not measure coupling.

1005 The formal field predicate requires a jointly continuous kernel, a normal-
1006 ized substrate, coupling equal to the kernel average, and no single site pro-
1007 viding more than half that strength. Toy constructions satisfy it. A cortical
1008 realization must specify the scale at which the kernel and its reduction ap-
1009 ply: calcium imaging reports different state-dependent modularity patterns
1010 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

1011 Cortical columns are the obvious candidate unit, and they answer one
1012 of the two demands the formalism makes. As the oscillator index they are
1013 a serious proposal: they are delineated by response properties and anatomy
1014 rather than by the agreement they would be used to establish (Mountcastle,
1015 1997), and a phase can be estimated from a column-sized territory at the
1016 electrode spacings used for cortical wave measurements (Townsend et al.,
1017 2015). As the content cover they are a poor one, for three separate reasons.

1018 A parcellation into non-overlapping columns is a disjoint cover, which satisfies
1019 the compatibility obligation without constraining anything (Section 4). The
1020 cover condition additionally requires the patches to exhaust the substrate,
1021 while much of association cortex has no accepted columnar parcellation. And
1022 the column's standing as a canonical cortical unit is contested (Horton &
1023 Adams, 2005; Rakic, 2008). Offering columns for both roles at once therefore
1024 supplies the oscillator index and leaves the content cover to be identified on
1025 separate evidence, which is the bridge assumption E78 records.

1026 The spatial reduction is also testable. A row-sum aggregation at zero
1027 diagonal tracks the uniform-coupling control over the tested spatial decay
1028 range, while frequency heterogeneity displaces the operational threshold. The
1029 fit window biases both estimates relative to $2D$; agreement with the control
1030 is therefore the supported conclusion. Supplemental Material reports the
1031 reduction, its numerical error and the distinct assumptions needed for the
1032 kernel-average identification in Eq. (5).

1033 The field commitment requires physical evidence beyond a choice of math-
1034 ematical representation, and the available support comparisons do not supply
1035 it. A finite matrix spread over positive-mass cells preserves its total cou-
1036 pling and phase correlation exactly, while point-supported vanishing under
1037 an atomless measure and a disadvantage from excluding a better-correlated
1038 pair concern specified support constraints; Section 6 reads the same compar-
1039 isons against a digital candidate. Neither supplies energy prices.

1040 A spike-time shift alone supplies no coupling rate. The dimensionless

1041 product of population size N , timing sensitivity $\mathcal{T}$, field amplitude E and
 1042 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N \mathcal{T} E f. \quad (15)$$

1043 This rate describes how induced timing shifts accumulate into drift. Nei-
 1044 ther it nor the installation factor κ follows from a heat bound. Testing the
 1045 physical threshold requires independent estimates of the interaction kernel,
 1046 its aggregation and diffusion in the same cortical regime.

1047 **9.2 Extracellular geometry and recovery**

1048 The proposed control variable is extracellular geometry. Extracellular vol-
 1049 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &
 1050 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
 1051 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively
 1052 at $0.3\text{--}0.6 \text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,
 1053 an effective-medium description predicts that a larger extracellular fraction
 1054 raises conductivity and reduces field amplitude.

1055 The coupling-gated hypothesis consequently predicts that extracellular
 1056 contraction on awakening raises E and, holding the other factors in Eq. (15)
 1057 fixed, raises effective coupling. This argument supplies a direction; the
 1058 geometry–coupling relation, its magnitude and its role in human recovery
 1059 require measurement.

1060 For a smooth crossing at nonzero speed, evaluating the stationary branch
 1061 along the changing control gives the benchmark

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (16)$$

1062 This follows from Eq. (3) at fixed D and concerns the stationary branch
 1063 evaluated at $K(t)$. Reading it as observed recovery requires an indepen-
 1064 dently justified tracking approximation. The crossing controls in Section 8.2
 1065 leave that approximation unsupported at onset, even though a separation of
 1066 geometry and phase time scales can support it away from threshold. The
 1067 awakening hypothesis therefore needs a calibrated dynamical prediction for
 1068 an operational onset above the finite-size fluctuation floor. Crossing K_c alone
 1069 does not supply a recovery latency.

1070 **9.3 A protocol for the awakening experiment**

1071 The physical test has six requirements:

- 1072 **1. Coupling and diffusion in the same units.** Estimate them from re-
 1073 solved phase dynamics, separating heterogeneity and observation noise.
 1074 Measure the drive or perturb the system to distinguish endogenous cou-
 1075 pling.
- 1076 **2. Geometry calibrated to coupling.** Validate the geometry observ-
 1077 able and its relation to effective coupling, with uncertainty. The con-

ductivity argument supplies only the expected direction.

3. **Calibrated phase measurements.** Estimate r by resultant length and a by unweighted regression of log density on centered cosine. Choose site count and binning from a dependence-aware calibration over the intended concentration range. The tested designs require thousands of sites at the observed concentrations; temporal pooling changes the spatial estimand.

4. **An independent recovery endpoint.** Measure behavioural recovery separately from phase organization.

5. **A predeclared dynamical onset criterion.** Set a coherence threshold above the finite-size floor and evaluate its latency under the measured coupling trajectory, including ramp speed or fluctuations. Record a spatial observable beside the resultant, since a winding field reports a low resultant at coupling that sustains local order (Section 8.3).

6. **Matched model comparison.** Compare coupling increase, diffusion change, common drive and alternative recovery models under the same observation and filtering assumptions.

Failure of the concentration–coherence relation over an informative range constrains the density model. Failure of the calibrated geometry–coupling association constrains the physical gate. Better prediction of independent recovery by a competing mechanism counts against its proposed role. A

1099 favorable outcome still leaves the content of coordinated activity to be ex-
1100 plained.

1101 10 Discussion

1102 Two things have been established here, and they are of different kinds. The
1103 constraints of Section 2 hold outright: synchrony does not deliver compatible
1104 content, and where shared encoders let it bound content at all, the bound
1105 that holds uniformly over pairs does so only inside patches too small to
1106 span the territories at issue, and distance costs either a connected cover or
1107 a locked configuration with a declared spectral gap. The resultant does not
1108 determine the state; a unique fixed point is not a reconstruction; a region that
1109 cannot resolve a family cannot encode it; and agreement about a change takes
1110 communication. Any account that reads unity off phase coherence has to
1111 meet them, this one included. The account set out in Section 1 is conditional
1112 at every step by contrast: each connection in it is a hypothesis supplied
1113 rather than a result derived, and Table 1 names all eight. Its four non-
1114 standard ingredients, named in Section 1, are the installed-energy condition,
1115 the patch-local form of the content bound, the criterion separating a winding
1116 field from weak coupling, and the machine-checked composition with its joint
1117 witness. What follows takes the conditions in turn, as commitments the
1118 proposal incurs, as measurements that would bear on them, and as questions
1119 the framework leaves open.

1120 10.1 Unity and a minimal self as empirical commit- 1121 ments

1122 We propose the state glued from compatible local descriptions as a corre-
1123 late of conscious unity, and its accurate internal reconstruction as a correlate
1124 of a minimal self. The motivations are that different contents belong to
1125 one experienced situation and that the situation is experienced from a per-
1126 spective. The conditional results identify relations a realization would need:
1127 sufficient installed resources for the specified phase model, agreement about
1128 shared content, and an encoding that supports reconstruction at the required
1129 fidelity. Discharging E12–E89 for cortex would make the composition a state-
1130 ment about cortex: that a cortical field above threshold relaxes to a cover
1131 whose compatible local descriptions glue, and that the glued state is the
1132 unique fixed point of a reconstruction map realized there. It would leave the
1133 two identifications above exactly where they are, because neither is among
1134 the eight and neither follows from their conjunction; they are the commit-
1135 ment this subsection names, and Section 10.2 lists the three observations
1136 that would count against it.

1137 Compatibility allows different descriptions to contribute different infor-
1138 mation. It also permits uncertainty and conflicting judgments: apparent cup
1139 position and judged cup position can be distinct quantities in a consistent
1140 description of their discrepancy. Which quantities are shared must be de-
1141 termined independently. Choosing them solely to secure agreement would

1142 deprive the proposal of empirical force.

1143 The minimal-self proposal concerns the perceiver’s position and condition
1144 within the situation, such as the cup’s location relative to a body that can
1145 reach it. The encoding could be implicit and need carry no autobiograph-
1146 ical narrative. Its proposed necessity is philosophical: even an unreflective
1147 episode has a perspective. Gluing by itself requires no self-encoding, and re-
1148 constructability alone supplies no functional role. Altering the encoded hand
1149 or viewpoint while holding the external scene fixed should change estimates
1150 of the perceiver’s relation to the cup as the readout predicts. Whether that
1151 represented perspective is experienced remains a further question.

1152 **10.2 What a test of the full proposal must establish**

1153 Both uniqueness results concern specified objects. They choose neither the
1154 cover nor the encoding, and do not select between candidate subjects. In-
1155 tegrated information theory treats exclusion explicitly (Tononi et al., 2016);
1156 here selection requires independent evidence about neural organization. A
1157 constant readout can contract and fix a predetermined state while discard-
1158 ing its input. The family criterion in Section 5.2 tests the sensitivity that
1159 this construction lacks, provided the relevant states and fidelity are declared
1160 independently.

1161 A test must separately measure coupling relative to diffusion, content
1162 compatibility and self-reconstruction, alongside an independent endpoint for
1163 experience. Meeting all three without a unified episode would challenge suffi-

1164 ciency. A unified episode with incompatible descriptions would challenge the
1165 unity identification; one lacking the specified self-encoding would challenge
1166 the reflexive requirement. Awakening, dreaming and dissociative states offer
1167 possible settings, but absence of report cannot by itself define absence of
1168 experience.

1169 The next physical requirement is a calibrated geometry–coupling relation
1170 and resolved phase dynamics. The installation bound contributes a nec-
1171 essary resource condition where its hardware representation applies; cortical
1172 mode prices and field-mediated coupling require measurement. For the repre-
1173 sentational connection, an informative experiment asks whether a controlled
1174 change in coordination alters independently decoded content agreement, with
1175 controls for sensory information and decoder quality. The constructed-data
1176 estimator establishes feasibility for specified contents. Neural local descrip-
1177 tions, shared overlap encoders and the self-encoding have yet to be identi-
1178 fied. The population-size dependence of the phase-to-content bound limits
1179 its quantitative use in the form Section 4.2 reads off the sheet, and each
1180 way around it asks the measurement for something further: a located excep-
1181 tional set, before a count of disagreeing pairs bears on a named overlap; a
1182 cover whose nerve is connected and whose patch diameters are measured; or
1183 a frequency-locked population over a coupling graph whose connectivity is
1184 measured structurally rather than estimated from the activity the bound is
1185 about.

1186 The separation results explain why these measurements must be inde-

pendent. A stationary distribution leaves several recovery mechanisms indistinguishable. Equal phase order can accompany incompatible descriptions or different task information. The tested plasticity rule preserves coherence while losing its designated template. Each finding concerns the specified model and observable; their role is to delimit the evidence needed for the proposed connections.

10.3 Explanatory scope

Integrated-information, global-workspace and predictive-processing accounts organize unity, access and modelling differently (Tononi et al., 2016; Baars, 1997; Dehaene et al., 2014; Friston, 2010). The present proposal contributes a candidate physical interaction and conditional relations between local content and self-encoding. Gluing and reconstruction permit multiple substrates; the electromagnetic commitment enters through physical calibration. That permission is wider than the unconditional results of Section 2 alone. The conditional chain is substrate-neutral in the same way: its field condition asks for a jointly continuous kernel on a normalized substrate, a coupling equal to that kernel’s double average, a noise that is the field’s own, and no single site carrying more than half the coupling—and none of these distinguishes an electromagnetic kernel from any other continuous mean-field kernel. The site condition is the only one with physical flavour, and it holds vacuously wherever the substrate has no atoms (`no_site_dominates`), which is the continuum case the electromagnetic hypothesis is about. The chain

1209 therefore transfers to extracellular ion diffusion, to gap junctions modelled
1210 as a density, or to a mechanical or optical medium, modulo the calibration of
1211 Section 9.1. What carries the electromagnetic identification is that calibra-
1212 tion together with the extracellular-geometry argument of Section 9, which is
1213 the one place a field equation enters. The formalization checks implications
1214 under their hypotheses, and its witnesses establish realizability on declared
1215 models. Two things stand outside that result and are not supplied by it: the
1216 eight hypotheses are discharged for toy components rather than for cortex,
1217 and the identification of the glued state with unity and of its reconstruction
1218 with a minimal self is a further commitment that the eight do not contain.

1219 The explanatory question is whether the specified relations occur in neural
1220 activity and matter for experience. In the cup example, this means establish-
1221 ing how coordination helps maintain one represented situation and how the
1222 perceiver’s place within it contributes to the episode. The proposal identi-
1223 fies candidate correlates and tests of that identification. Why those relations
1224 would be accompanied by experience remains unexplained.

1225 Notation

Table 2: Symbols used throughout, with the section that defines each. Where one letter carries more than one meaning, the meanings are separated by the section they are read in.

Symbol	Defined	Meaning
θ_i, ω_i	Eq. (1)	Phase and natural frequency of oscillator i

Table 2 (continued)

Symbol	Defined	Meaning
N	Eq. (1)	Number of oscillators in the population the quantity is read on
D	Eq. (1)	Phase diffusion, in inverse time
K_{ij}, K	Eq. (1)	Pairwise and scalar mean-field coupling, $K_{ij} = K/N$
K_c	Eq. (2)	Synchronization threshold, $K_c = 2D$
r, r_N	Eq. (1)	Resultant length (order parameter), on the population it is read on
a, ψ	Eq. (2)	Concentration $a = Kr/D$ and collective phase of the stationary density
q	Section 8.3	Winding number of a phase field read around a ring
μ	Eq. (4)	Substrate measure, normalized to one
Ξ, ϕ_i, u_i, c_i	Eq. (4)	Actuator configuration; mode profile, installation price and occupancy of mode i
U_{inst}, κ	Eq. (5)	Installed energy and the mode-data conversion factor bounding $K_{\text{mf}} \leq \kappa U_{\text{inst}}$
L	Section 4.2	Lipschitz constant of an encoder or readout
ε, δ	Section 4.2	Content and observation discrepancy tolerances
M, b	Eq. (10)	Size of a declared family of states and code length in bits
Λ, τ	Eq. (8)	Contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ and its relaxation time
T	Section 5	Number of communication rounds to a deadline
ν	Section 6	Fan-in of a communication graph: the most sites any one site hears from
$k_B T, B$	Eq. (14)	Thermal scale, read as one symbol, and heat budget of a register
$\gamma, \mathcal{T}, f$	Eq. (15)	Field-to-coupling conversion rate, timing sensitivity and frequency
α	Section 9	Extracellular volume fraction
λ	§8.3, §9	Eigenvalue the coupling kernel supplies at a winding; extracellular tortuosity
Ω, λ_2	Section 4.2	Common rotation rate of a frequency-locked configuration, and the algebraic connectivity of the coupling graph

Table 2 (continued)

Symbol	Defined	Meaning
β, k	Section 4.2	Chord diameter of a patch and the diameter of a cover’s nerve, in hops
$\mathcal{C}, e_{i,x}$	Section 4.2	Content space and the phase encoder patch i reads at site x
E, R	Eq. (10)	Encoder and readout of the reconstruction criterion
$\mathcal{A}, F$	Eq. (8)	Encoding region and the reconstruction map $F(s) = \text{readout}(s \mathcal{A})$
$\mathcal{E}$	Eq. (13)	Energy observable shared by the stages of a continuing protocol
z_i, Γ, A_{ij}	Section 8.3	Complex amplitude, linear gain and coupling matrix of the amplitude layer
$\Delta\theta_d$	Section 8.3	Phase a winding advances across separation d
E	Eq. (15)	Field amplitude, in the cortical identification
U	§3.2, §4	Stored energy $U(\Xi)$ of a configuration; a patch U of the cover
s, s_*	§4–5	A global section and the one the reconstruction fixes

Data and code availability

The Lean development, simulation scripts, tests and saved numerical summaries are maintained in the GitHub repository `bobaseb/coupling_gated_cortical_coherence`. Publication simulation values are generated from saved summaries without rerunning production sweeps. The exploratory EEG analysis uses OpenNeuro dataset ds005620 (Bajwa et al., 2025); pre-processing and selection are described in Supplemental Material. The repository contains the reproducible analysis and does not redistribute the raw

1234 recordings.

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1238 **Use of AI tools**

1239 AI assistance to this work was substantial and is not separable by task. Sev-
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1241 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
1242 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra; GPT-5.6 Terra,
1243 Luna and Sol), Claude Code (Claude Opus 5), Gemini Pro through its web in-
1244 terface, and Anthropic’s Fable and DeepSeek v4 Flash through OpenRouter,
1245 accessed with the Hermes CLI. Between them they contributed to the Lean
1246 development, the simulation and analysis code, and the drafting and revision
1247 of both documents. Because the tools were used in combination and in alter-
1248 nation, no section, proof, figure or result is attributable to any one of them,
1249 and none is claimed to be. The author is responsible for all content, including
1250 everything a tool produced. Machine checking establishes the stated Lean
1251 results under their hypotheses; it does not validate biological interpretations
1252 or replace scientific review of generated code and prose.

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